Amnesia by Design: Diagnosing Temporal Traceability in Transformer Representations

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Abstract. Words do not preserve fixed meanings across time. Without representations that reflect this, token@time queries become impossible: there is no way to ask where the meaning of *gay* stood in 1920, or how it moved by 1980, if both queries return the same geometric object. Yet time-conditioned transformers have never been tested on whether they actually achieve this geometrically. Knowing that a representation encodes the correct period label is not the same as knowing whether it occupies a period-appropriate neighborhood, a distinction we formalize as *temporal traceability*. We introduce a controlled diagnostic suite with planted trajectories (stable, drifting, bifurcating) to measure it directly. Three findings emerge. Any form of temporal conditioning more than doubles neighborhood displacement relative to the non-temporal baseline. However, a simple global period signal and joint token-time conditioning are geometrically equivalent under reliable local markers; the difference only appears when those markers are removed. Finally, for words that develop two coexisting senses, mean prototypes fail systematically: they collapse both senses into a centroid that represents neither. The results point to the need for training objectives aligned with trajectory geometry.

Keywords: Temporal semantics · Semantic change · Transformer models · Traceability · Diagnostic evaluation

1 Introduction

Words do not preserve fixed meanings across time. In 1900, *broadcast* referred to scattering seeds across a field; today it describes a social media post reaching millions. The computational study of such shifts, lexical semantic change detection (LSCD), typically works by training distributional representations on time-sliced corpora and comparing them across periods [8, 9]. This approach has produced important methods and benchmarks, but it summarizes change as a single similarity score between two separately trained models: `broadcast@1900` and `broadcast@2020` are implicit in those models, not directly inspectable within a shared geometric space.

We call a representation *temporally traceable* if the same surface token, queried at different periods, occupies neighborhoods consistent with its meaning at that period—making `token@time` a directly inspectable object. Consider *gay*, which in the early twentieth century was mainly associated with *cheerful* and *merry*, and by the late twentieth century with *queer* and *lesbian* [5]. Whether this shift involved movement toward or away from stigmatized terms [5] is an empirical question that requires `gay@1920` and `gay@1980` as directly comparable objects.

Testing whether a representation achieves this property requires a controlled environment where the correct answer is known. Natural-language corpora entangle semantic change with frequency, genre, and subcorpus shifts, making it hard to attribute a representational effect to architecture rather than data. We therefore introduce a diagnostic suite with planted trajectories—stable, drifting, and bifurcating—and a battery that separates neighborhood geometry from period-label recoverability.

Against this suite we evaluate one non-temporal baseline (Standard) and three temporal conditioning architectures: Additive (global period shift), Token-Time (joint token-period projection), and Memory-Augmented (causal prototype history). Our research question is whether any form of conditioning improves geometric traceability over the non-temporal baseline, and what each design contributes. Evaluation on noisier regimes and natural corpora is left for future work.

The paper makes four contributions. (1) We establish temporal traceability as a geometric property distinct from period-label recoverability, and motivate why standard training objectives do not directly optimize it. (2) We introduce a controlled diagnostic suite with planted trajectories — stable, drifting, and bifurcating — and a four-diagnostic battery that targets neighborhood geometry rather than label recovery. (3) Using this suite, we show that all three conditioning architectures more than double neighborhood displacement over the non-temporal baseline and are geometrically equivalent under reliable local markers; joint token-period projection pulls ahead only when markers are removed. (4) We identify mean-prototype aggregation as a failure mode for bifurcating subjects: when a word develops two coexisting senses, a mean prototype collapses both into a centroid that represents neither.

2 Related Work

Lexical semantic change detection (LSCD) is the dominant computational framing for meaning shift. Diachronic word embeddings [5] learn one distributional space per time slice, align independently trained models, and measure movement in neighborhood structure. We share the intuition that semantic change is geometric, but not the architecture: LSCD yields a scalar comparison between spaces rather than a jointly parameterized `token@time` representation. Standard benchmarks formalize this task [8, 9], while also inheriting natural-corpus confounds such as frequency and genre drift; shared tasks for Spanish show that manual annotation is required even for modern corpora [11]. Our controlled diagnostic suite plants trajectories so that traceability is attributed to model design, not corpus composition.

Temporal neural language models avoid separate spaces by conditioning one model on time. Prior work uses timestamps for knowledge-intensive QA, masked language modeling, or period-stratified downstream evaluation [3, 7, 6]. These results show that temporal conditioning can improve period-appropriate prediction but do not test whether the same token occupies different representational neighborhoods at different periods, which is the geometric question this work addresses.

Finally, semantic change may involve bifurcation rather than monotonic drift. Giulianelli et al. show that words can develop coexisting senses [4]; mean-pooled representations then collapse both senses into a centroid that represents neither. Our stable, drifting, and bifurcating trajectories test this distinction directly, and the memory extension exposes where mean-pooled prototypes fail.

3 Controlled Diagnostic Suite

The diagnostic suite gives each subject a known temporal trajectory, making it possible to ask directly whether a model's representation of a subject at $t0$ has moved toward the expected neighborhood by $t9$. This distinguishes it from natural-language LSCD benchmarks, where ground truth is often annotated or inferred post-hoc and corpus confounds are hard to separate from genuine semantic change.

The corpus uses a subject–verb–object (SVO) grammar: every sentence consists of one subject, one verb, and one object. Two semantic neighborhoods, N_1 and N_2, partition the vocabulary of 8 verbs and 8 objects: N_1 uses $\{V1, \ldots, V4\}$ and $\{O1, \ldots, O4\}$; N_2 uses $\{V5, \ldots, V8\}$ and $\{O5, \ldots, O8\}$. A sentence generated from N_1 therefore looks like `S5 V2 O3`, while one from N_2 looks like `S5 V6 O8`; the subject token is the same in both cases, but its co-occurring words differ. Intuitively, the intended meaning of subject s at period t is how often it appears in N_1 sentences; formally, $P(N_1 \mid s, t)$ is the probability that a sentence is drawn from N_1. The vocabulary contains 30 subjects; each subject's trajectory is determined by how $P(N_1 \mid s, t)$ evolves across the 10 periods $t0$ through $t9$.

Figure 1 (center) shows the three planted trajectory classes. Stable subjects maintain an approximately constant $P(N_1 \mid s, t)$ throughout. Drift subjects follow a monotonic trajectory from near-exclusive N_1 ($P(N_1 \mid s, t) \approx 1$ at $t0$) to near-exclusive N_2 ($P(N_1 \mid s, t) \approx 0$ at $t9$): a drifting subject that appears in sentences like `S5 V2 O3` early on will appear in sentences like `S5 V6 O8` by the final period. Bifurcating subjects begin in N_1 and move toward a mixed state ($P(N_1 \mid s, t) \approx 0.5$) where both neighborhoods appear with equal frequency, approximating two coexisting senses. The suite contains 10 subjects per class.

Each local marker is drawn from the opposite neighborhood 25% of the time (Figure 1, left), so the model cannot recover the planted neighborhood from co-occurring words alone; it must use subject identity together with the period. The `true_context` label is withheld from training and used only for evaluation.

The controlled diagnostic suite isolates architectural effects that natural corpora confound with frequency, genre, and composition shifts. By planting trajec-

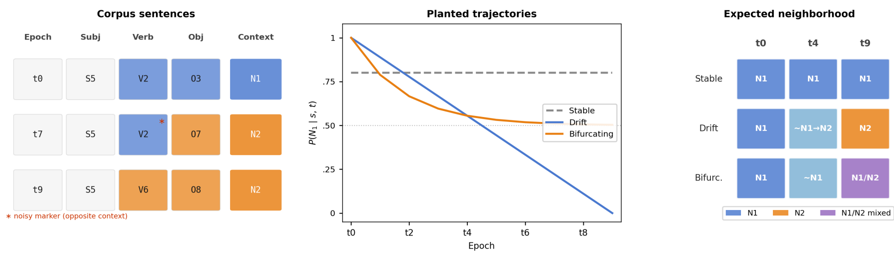

Fig. 1. Controlled diagnostic suite overview. *Left*: SVO sentences for a drifting subject across three periods; an N_1 verb appearing in an N_2 sentence at $t7$ (marked $*$) illustrates the deliberate noise in local markers. *Center*: planted $P(N_1 \mid s, t)$ trajectories for Stable (flat), Drift (monotonic), and Bifurcating (levelling at 0.5) subjects across $t0$–$t9$. *Right*: expected nearest-neighbor behavior for a traceable representation.

tories with known $P(N_1 \mid s, t)$, we can ask directly whether a model's geometry matches the generating process. The scope of these controlled diagnostics is revisited in the Discussion.

The diagnostic suite supports several evaluation conditions. The standard split uses the same 75% marker accuracy as training. Two harder splits force the verb, or both verb and object, to come from the opposite context. An ambiguous split sets fidelity to 50%, so local markers carry no reliable signal; only subject identity and period remain. Two additional conditions test temporal behaviour and are described in Section 5: the contrastive set presents the same sentence under two different period labels to check whether changing only the period changes the preferred context; the continuation set uses periods $t8$–$t9$, held out from MLM training (which covers $t0$–$t7$), to test generalisation.

4 Temporal Conditioning Architectures

We evaluate one non-temporal baseline and three temporal conditioning architectures, all sharing the same Transformer encoder backbone. Standard (no period input) is the non-temporal baseline. The three conditioning architectures differ in how they introduce the period signal: Additive applies a global period shift to every token; Token-Time uses a joint token-period projection; and Memory-Augmented extends Token-Time with causal prototype history.

All models follow the same outer pipeline based on the Transformer encoder [10]. A sentence is tokenized as `[CLS] S V O [SEP]`. Each token w_i is mapped to an input embedding $z_i \in \mathbb{R}^d$. The sequence $(z_1, \ldots, z_n)$ is processed by a Transformer encoder, and the final hidden state at the subject position,

$$(h_1, \ldots, h_n) = \mathrm{Encoder}(z_1, \ldots, z_n), \qquad h_s = h_{\mathrm{subj}},$$

is the `token@time` representation inspected by the diagnostics. All models are trained with masked language modeling (MLM, [2]): some tokens are masked,

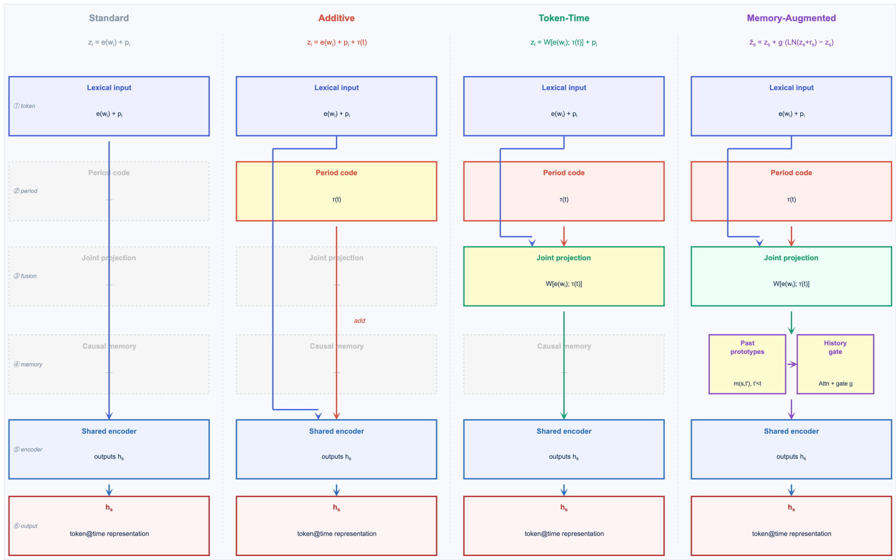

Fig. 2. Controlled design comparison. **Standard**: Transformer baseline with no temporal signal. **Additive**: temporal baseline that adds the same period vector to every token. **Token-Time**: Token-Time Transformer using joint token-time projection. **Memory-Augmented**: extension of Token-Time with Prototype Memory and Temporal Attention. New components are highlighted in yellow; dashed grey boxes mark inactive components.

and the model reconstructs them from the remaining tokens and, when available, the period input. The variants differ only in how they build z_s before the encoder.

The temporally informed models use a shared period vector $\tau(t) \in \mathbb{R}^d$, a learned projection of sinusoidal features of t/T, where T is the number of training periods. It is distinct from the position encoding p_i, which marks subject, verb, or object position. Additive and Token-Time differ in how they combine $\tau(t)$ with token identity; Memory-Augmented then adds causal memory. Figure 2 summarizes the comparison.

4.1 Baseline: Standard Transformer

Standard is the null hypothesis: no period signal is available, so any temporal behavior must arise from changes in the observed words alone.

$$z_i = e(w_i) + p_i.$$

4.2 Additive Time-Conditioned Transformer

Additive tests the simplest form of temporal conditioning: a uniform global period signal. It adds the same period vector $\tau(t)$ to every token:

$$z_i = e(w_i) + p_i + \tau(t).$$

Now `S5 V2 O3` at $t0$ and $t9$ are distinguishable: the subject receives $\tau(0)$ or $\tau(9)$. The same shift is applied to every token, so Additive tells the whole sentence which period it belongs to but does not bind period to individual token identities.

4.3 Token-Time Transformer

The Token-Time Transformer in Figure 2 asks whether exposing token identity and period jointly before self-attention is enough. Instead of adding the same period vector to all tokens, Token-Time concatenates each token embedding with $\tau(t)$ and projects the pair:

$$z_i = W[e(w_i);\, \tau(t)] + p_i,$$

where $e(w_i), \tau(t) \in \mathbb{R}^d$ and $W \in \mathbb{R}^{d\times 2d}$. Although W is linear, it exposes token-period inputs before self-attention and later nonlinear layers. Thus `S5` at $t0$ and `S5` at $t9$ enter the encoder as distinct period-conditioned inputs.

4.4 Memory-Augmented Extension

The Memory-Augmented extension asks whether causal history helps beyond token-time conditioning. It starts exactly like Token-Time: for a current sentence containing `S5` at $t7$, the model first builds token-time input embeddings for `[CLS] S5 V O [SEP]`. It then modifies only the subject embedding before the Transformer encoder sees the sentence.

The additional information comes from Prototype Memory. The extension stores one prototype vector per subject and past period. A prototype $m(s, t)$ is the average representation of subject s across training sentences from period t. The memory is causal: at $t7$, the model may consult $\{m(\texttt{S5}, 0), \ldots, m(\texttt{S5}, 6)\}$, but not prototypes from $t7$ or later.

Temporal attention turns this list of past prototypes into a history summary for the current subject. Let z_s be the current Token-Time subject embedding and let M_s be the matrix of available causal prototypes, one row per past period. The attention operation uses z_s as the query and M_s as keys and values:

$$r_s = \mathrm{Attn}(z_s, M_s, M_s).$$

The vector r_s summarizes the parts of the subject's history that are most relevant to the current period. A learned scalar gate g then decides how much this history summary should change the current subject embedding:

$$\tilde{z}_s = z_s + g \cdot (\mathrm{LN}(z_s + r_s) - z_s)\,.$$

Here, LN denotes layer normalization, and g is a learned scalar gate initialized at zero so that the extension reduces to Token-Time at initialization. Finally, $\tilde{z}_s$

replaces z_s in the input sequence; verb and object embeddings are unchanged, so the subject is the only token that enters the encoder informed by causal prototype history.

5 Evaluation Protocol and Diagnostics

All models in the comparison chain are trained with masked language modeling over the synthetic SVO corpus. The planted `true_context` label is withheld from training and used exclusively for post-hoc evaluation: it tests whether the learned subject representation reflects the planted temporal trajectory. The primary evaluation target is h_s, the final hidden state at the subject position, because subjects are the lexical items whose meanings are designed to drift, remain stable, or bifurcate over time.

We use four diagnostics, each targeting a distinct aspect of temporal representation.

D1: Linear probe [1]. Asks whether h_s encodes the planted context label at all. The encoder is frozen; a linear classifier is trained on h_s representations extracted from the held-out test sentences of the standard (75%-fidelity) split and evaluated on the ambiguous and continuation splits. High accuracy indicates that period-appropriate context is linearly separable in the representation, but does not reveal whether that separation shifts systematically over time.

D2: Context drift score. Asks whether the neighborhood of a subject shifts in the expected direction across periods. For each period and subject class, we retrieve the $k = 10$ nearest neighbors of h_s by cosine similarity (excluding the query itself) and record the proportion assigned to N_1. A representation is traceable at period t if this proportion tracks $P(N_1 \mid s, t)$, the same probability that governs corpus generation at that period. For drifting subjects, the proportion should therefore decrease monotonically from $t0$ to $t9$; for stable subjects it should remain near its initial value; for bifurcating subjects it should decrease less steeply, as the representation splits between neighborhoods rather than migrating from one to the other.

D3: Contrastive diagnostic. Asks whether the period label alone drives representational change. We present the same sentence with two different period labels and compute the sign-flip rate: the proportion of pairs for which swapping the period reverses the nearest-neighbor assignment, from N_1-dominant to N_2-dominant or vice versa. A model that ignores the period input produces a sign-flip rate of zero; a model that responds to it produces a rate above zero.

D4: Continuation diagnostic. Asks whether the learned temporal mapping generalizes beyond training. We evaluate on held-out periods $t8$–$t9$, excluded from MLM training, and operationalize the diagnostic as linear-probe accuracy on representations from these periods. A model that has internalized the direction of change, rather than memorizing period-specific patterns, should preserve context recoverability in unseen periods.

The four diagnostics are interpreted through a fixed comparison chain. All three conditioning architectures vs. Standard isolates the effect of adding any temporal signal. Additive vs. Token-Time compares the two simpler designs: global shift versus joint token-period projection; in this regime the difference is visible only on the ambiguous-split probe. Memory-Augmented vs. Token-Time tests the contribution of causal prototype history, the mechanism most expected to affect D3 and D4.

For the Memory-Augmented extension, we include an *oracle-memory* diagnostic: learned prototypes are replaced by fixed vectors encoding the true historical $P(N_1 \mid s, t)$ trajectory, with the rest of the architecture unchanged. This isolates whether the failure of learned prototypes is due to the memory pathway itself or to the aggregation mechanism; improvement under oracle prototypes indicates that the pathway can use better historical signals, while learned mean prototypes remain the weak point.

6 Results

We report means and 95% confidence intervals over 31 training seeds. Conditioning of any form substantially moves drifting subjects relative to the non-temporal baseline: all three conditioned models reach $\Delta \approx -0.56$ vs. -0.221 for Standard, more than doubling the neighborhood displacement. Additive and Token-Time are indistinguishable on aggregate drift in this controlled setting. Memory-Augmented does not improve over Token-Time.

Implementation details. All models share the same Transformer encoder: $d = 64$, 4 attention heads, 2 layers, feed-forward width 128, dropout 0.1. The period encoding projects 32 sinusoidal features of t/T to d via a two-layer MLP. Training uses AdamW (lr $= 10^{-3}$, weight decay $= 10^{-2}$) with a cosine annealing schedule, 30 epochs, batch size 64. The corpus contains 5,000 SVO sentences; MLM training uses 3,159 sentences from periods $t0$–$t7$; 763 sentences from the same periods form the held-out evaluation set; 794 sentences from $t8$–$t9$ are reserved for the continuation diagnostic.

6.1 Temporal Neighborhood Movement

Figure 3 and Table 1 show the context drift score across periods. Standard reaches $\Delta = -0.221$ (95% CI $[-0.240, -0.203]$); Token-Time reaches $\Delta = -0.563$, more than doubling the displacement. Additive reaches an equivalent value ($\Delta = -0.567$), confirming that a global temporal signal reproduces aggregate drift in this high-fidelity setting.

The class-specific curves serve as controls (Table 1). Stable subjects show smaller movement in absolute magnitude (Δ between -0.150 and -0.180 for conditioned models, versus $\Delta \approx -0.56$ for drifting subjects), but the direction is consistently negative. This reflects a global displacement of the embedding space induced by the period vector: even subjects with flat planted trajectories are pulled along the learned temporal direction, at lower magnitude. Bifurcating subjects show intermediate drift: conditioned models reach $t9$ values near 0.53,

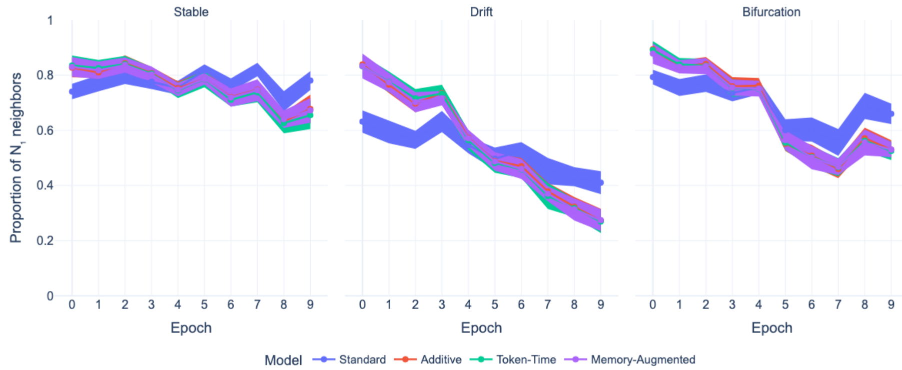

Fig. 3. Context drift score across periods. All conditioned models move drifting subjects from N_1 toward N_2 more than Standard; Additive, Token-Time, and Memory-Augmented produce similar trajectories. Stable and bifurcating subjects provide class-specific controls.

Table 1. Context drift score at the first and last periods. Values are means over 31 seeds; $\Delta = t9 - t0$; 95% CI half-widths in brackets.

Class	Model	t0	t9	Δ [CI$_\pm$]
Stable	Standard	0.741	0.781	+0.040 [±0.017]
	Additive	0.827	0.678	−0.150 [±0.027]
	Token-Time	0.835	0.655	−0.180 [±0.024]
	Mem-Aug.	0.830	0.672	−0.158 [±0.021]
Drift	Standard	0.632	0.410	−0.221 [±0.018]
	Additive	0.839	0.272	−0.567 [±0.019]
	Token-Time	0.833	0.269	−0.563 [±0.020]
	Mem-Aug.	0.833	0.274	−0.559 [±0.022]
Bifurc.	Standard	0.793	0.660	−0.133 [±0.018]
	Additive	0.895	0.530	−0.365 [±0.017]
	Token-Time	0.891	0.525	−0.366 [±0.016]
	Mem-Aug.	0.878	0.530	−0.348 [±0.015]

matching the planted endpoint $P(N_1 \mid s, t9) \approx 0.5$ for the mixed state. The geometric diagnostic distinguishes all three trajectory classes without relying on probe accuracy alone.

Paired equivalence testing (TOST, $\delta = 0.05$) formalizes this: diff$= -0.003$, 90% CI $[-0.026, +0.019]$, $p_{\mathrm{TOST}} < 0.001$; equivalence holds also at $\delta = 0.034$ ($p_{\mathrm{TOST}} = 0.014$). D2 thus confirms the geometric effect of joint conditioning; it is not a comparison between conditioning strategies.

6.2 Probe and Contrastive Diagnostics

The linear probe results reveal a distinction that drift score does not. On the standard (75%-fidelity) split, where local markers carry reliable context signal, all conditioned models outperform Standard (0.875 [±0.005]): Additive reaches 0.892 [±0.003], Token-Time 0.881 [±0.005], and Mem-Aug. 0.881 [±0.006]; on

Table 2. Main traceability diagnostics (means, 31 seeds). Drift: context drift score Δ for drifting subjects; Flip: contrastive sign-flip rate (D3); Ambig./Cont.: linear-probe accuracy on h_s. 95% CI in brackets.

Model	Drift	Flip	Ambig.	Cont.
Standard	−0.221 [±0.018]	0.000	0.565 [±0.004]	0.730 [±0.004]
Additive	−0.567 [±0.019]	0.602 [±0.053]	0.579 [±0.005]	0.763 [±0.003]
Token-Time	−0.563 [±0.020]	0.624 [±0.048]	0.595 [±0.006]	0.763 [±0.004]
Mem-Aug.	−0.559 [±0.022]	0.587 [±0.051]	0.598 [±0.008]	0.755 [±0.005]

this split Additive is above Token-Time. The ordering reverses on the ambiguous split, where local verb and object markers are uninformative: Standard reaches 0.565 $[0.561, 0.568]$, Additive 0.579 $[0.575, 0.584]$, Token-Time 0.595 $[0.589, 0.601]$, and Memory-Augmented 0.598 $[0.590, 0.606]$. The confidence intervals for Additive and Token-Time do not overlap, showing a reproducible advantage for joint token-time conditioning when only subject identity and period remain as evidence. On the continuation split, Additive and Token-Time reach the same accuracy (0.763; Table 2), matching their overlap on drift score. Probe accuracy alone is insufficient to establish temporal traceability because it measures recoverability of the planted label, not neighborhood geometry. Even so, the split-dependent ordering is preliminary evidence that the D2 equivalence is regime-specific: joint conditioning encodes temporal information more precisely when local markers are removed.

The contrastive sign-flip rates are similar across conditioned models: Standard 0.000, Additive 0.602 $[0.549, 0.655]$, Token-Time 0.624 $[0.576, 0.673]$, Mem-Aug. 0.587 $[0.536, 0.638]$. Confidence intervals overlap, so the contrastive diagnostic does not separate the conditioned models.

6.3 Memory Diagnostics

The Memory-Augmented extension does not improve over Token-Time. Across 31 seeds, learned prototypes reach a continuation accuracy of 0.755 $[0.750, 0.760]$, marginally below Token-Time at 0.763 $[0.759, 0.766]$ ($\Delta = -0.008$). The oracle diagnostic shows the limits of this architecture: replacing learned prototypes with fixed vectors encoding the true $P(N_1 \mid s, t)$ trajectory yields a continuation of 0.769 $[0.761, 0.777]$ and a sign-flip rate of 0.694 $[0.576, 0.812]$, versus 0.624 for Token-Time. The sign-flip difference ($\Delta = +0.070$) exceeds the continuation improvement ($\Delta = +0.007$), but both CIs are wide. The MLM objective optimizes token prediction, not trajectory geometry. Perfect historical prototypes can improve contrastive behavior, which depends on local period sensitivity, but they do not replace a training signal for geometric continuity across unseen periods.

7 Discussion and Conclusion

This work establishes that temporal conditioning makes `token@time` queries geometrically meaningful. Any form of conditioning more than doubles neighborhood displacement for drifting subjects relative to the non-temporal baseline,

Table 3. Memory diagnostics (means, 31 seeds). Continuation (D4): linear-probe accuracy on held-out periods $t8$–$t9$; Flip: contrastive sign-flip rate (D3); 95% CI half-widths in brackets.

Model	Cont.	Flip	Δ cont. vs. Token-Time
Token-Time (ref.)	0.763 [±0.004]	0.624 [±0.048]	–
Mem-Aug. learned	0.755 [±0.005]	0.587 [±0.051]	−0.008 [±0.006]
Mem-Aug. oracle	0.769 [±0.004]	0.694 [±0.059]	+0.007 [±0.007]

and the diagnostic suite distinguishes all three trajectory classes without relying on probe accuracy alone.

Within this controlled setting, the three conditioning architectures converge on aggregate neighborhood movement under reliable local markers: Additive and Token-Time are equivalent on drift score (TOST, $\delta = 0.05$, $p_{\text{TOST}} < 0.001$), showing that a global period shift and a joint token-period projection produce indistinguishable geometry in this regime. The only reproducible distinction appears on the ambiguous split (0.595 vs. 0.579, non-overlapping CIs), where local markers carry no signal, suggesting that token-level binding encodes temporal information more precisely when context is degraded. Layer-wise CKA corroborates this convergence: Additive vs. Token-Time similarity increases from 0.726 to 0.887 across encoder depth (Standard stays near 0.22), suggesting self-attention absorbs the input-level difference. Whether the designs diverge under noisier regimes and natural corpora remains open.

For bifurcating subjects, mean prototypes collapse two coexisting senses into a centroid that represents neither. The oracle diagnostic shows the memory pathway itself is sound: fixed vectors encoding the true $P(N_1 \mid s, t)$ trajectory increase the sign-flip rate by +0.070 over Token-Time (0.694 vs. 0.624). The bottleneck is aggregation, not the pathway; future work should explore memory mechanisms that preserve distributional structure rather than collapsing each period to a single prototype.

These conclusions come from controlled diagnostics, not open-domain diachronic language; the probe-level advantage of Token-Time over Additive (1.6 percentage points) does not appear on the drift score and may not generalize beyond fixed subject positions and a small vocabulary. The suite was co-designed with the evaluated architectures to characterize them, not as an independent validation. Token-Time layer-0 heads show greater temporal variance in attention mass (L0H3: 0.0075 vs. Additive 0.0033), yet ablating individual heads only partially reduces representational alignment (L0H0: $\Delta = -0.026$; L0H1: $\Delta = -0.021$), suggesting a distributed effect; a pilot noise sweep (3 seeds, slope$= +1.07$, $p = 0.002$) is consistent with Token-Time retaining a drift advantage under degraded markers, but replication is needed before stronger claims can be made.

References

1. Conneau, A., Kruszewski, G., Lample, G., Barrault, L., Baroni, M.: What you can cram into a single $&!#* vector: Probing sentence embeddings for linguistic properties. In: Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 2126–2136 (2018)
2. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: BERT: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL). pp. 4171–4186 (2019)
3. Dhingra, B., Cole, J.R., Eisenschlos, J.M., Gillick, D., Eisenstein, J., Cohen, W.W.: Time-aware language models as temporal knowledge bases. Transactions of the Association for Computational Linguistics (TACL) **10**, 257–273 (2022)
4. Giulianelli, M., Del Tredici, M., Fernández, R.: Analysing lexical semantic change with contextualized word representations. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 3960–3973 (2020)
5. Hamilton, W.L., Leskovec, J., Jurafsky, D.: Diachronic word embeddings reveal statistical laws of semantic change. In: Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 1489–1501 (2016)
6. Loureiro, D., Barbieri, F., Neves, L., Anke, L.E., Camacho-Collados, J.: TimeLMs: Diachronic language models from Twitter. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL): System Demonstrations. pp. 251–260 (2022)
7. Rosin, G., Radinsky, K.: Time masking for temporal language models. In: Proceedings of the 15th ACM International Conference on Web Search and Data Mining (WSDM). pp. 833–841 (2022)
8. Schlechtweg, D., McGillivray, B., Hengchen, S., Dubossarsky, H., Tahmasebi, N.: SemEval-2020 task 1: Unsupervised lexical semantic change detection. In: Proceedings of the 14th International Workshop on Semantic Evaluation (SemEval). pp. 1–23 (2020)
9. Tahmasebi, N., Borin, L., Jatowt, A.: Survey of computational approaches to lexical semantic change. Computational Approaches to Semantic Change pp. 1–91 (2021)
10. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention is all you need. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 30, pp. 5998–6008 (2017)
11. Zamora-Reina, F.D., Bravo-Marquez, F., Schlechtweg, D.: LSCDiscovery: A shared task on semantic change discovery and detection in Spanish. In: Proceedings of the 3rd Workshop on Computational Approaches to Historical Language Change (LChange). pp. 149–164 (2022)